What the numbers say

What does the data actually tell us, and what does it not?



Today

By the end of today, you will have

Produced three charts or tables that disaggregate or compare, each with an interpretation and a limitation.

- Compute and read shares, rates, averages and changes correctly.
- Disaggregate by place, group or time to find where the problem concentrates.
- Choose a chart that suits the comparison, and label it honestly.
- Separate what the numbers show from what they suggest.

Competencies: E4 · I3 · C5 · J3

Session plan

10 min	**Opening problem**
20 min	**Concept: reading numbers**
20 min	**Demonstration: Madhyapur data**
30 min	**Activity: data detective**
30 min	**Studio: own charts**
10 min	**Debrief and exit task**

Opening problem

- **Average separation across Madhyapur's wards rose from 24 to 31 per cent. Good news?**

What would you want to see before agreeing? (All Madhyapur figures are illustrative.)

Five numbers that are easy to misread

1. **Averages**

   An average can rise while most wards get worse, if a few improve a lot.

2. **Points and per cent**

   From 24 to 31 per cent is 7 points, or a 29 per cent rise. Say which.

3. **Rates and counts**

   More complaints may mean more problems, or more people able to complain.

4. **Missing data**

   Wards that did not report are often the worst ones.

5. **Denominators**

   Per household, per ward or per person: the choice changes the story.

Data detective: Madhyapur's wards (illustrative data)

Ward group	Households	Separation last year	Separation this year	Reporting
Planned colonies	40,000	35%	60%	All wards
Older core wards	60,000	22%	23%	All wards
Informal settlements	50,000	15%	14%	4 of 10 wards
New peripheral wards	30,000	28%	36%	All wards
City (weighted)	180,000	24%	31%	

Figures invented for teaching. The city average rose mainly because planned colonies improved; informal settlements barely reported.

Choosing a chart

Comparison	Use	Avoid
Across categories	A sorted bar chart	3D bars; pies with many slices
Over time	A line chart	Truncated axes that exaggerate change
Part of a whole	A stacked bar or a simple table	Pies with more than three parts
Two variables	A scatter plot	Implying causation from the pattern
A few precise values	A table	A chart for five numbers

Every chart needs a title that states the finding, labelled axes, the year and the source.

From a pattern to a claim

1

What the data show

Wards with twice-daily collection have higher separation.

2

What it might mean

Frequent collection makes separation easier, or richer wards get both.

3

What you can claim

The two go together. Frequent collection may help, but income could explain both. Treat it as a hypothesis.

Write moves together, is associated with or causes, and mean the difference (competency C5).

False precision

- **Separation rose by 6.94 per cent is less honest than separation rose by about seven points, on incomplete data.**

Round to what the data can bear. Give a range when sources disagree. Say what is missing.

Activity: Data detective

Find what a dataset can and cannot support, including the traps hidden in it.

Scenario

An illustrative district dataset for a school meal scheme: enrolment, meals served, days of operation and complaints by block for two years. It contains a misleading average, missing months, a change in counting method and a helpline effect.

What to do

1. Compute the district figure for each year, then the block figures.
2. Find where the average hides a different story.
3. Identify every missing or inconsistent value.
4. Write three observations, two interpretations and one limitation.
5. Present the claim you are most, and least, confident about.

Time 30 minutes

Groups Pods of four

Output Three observations, two interpretations, one limitation

Your case: Worksheet 5.2

Data analysis

1. Find the dataset that bears most directly on your priority hypothesis.
2. Disaggregate it by place, group or time.
3. Produce three charts or tables, each with a title that states the finding.
4. Write three observations, two interpretations and one limitation.
5. Record each source in the evidence register.

Hand in

Worksheet 5.2 by the start of Day 13.

Quality check

Three charts or tables, each disaggregated or comparative.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

What to take away

1. Disaggregate: averages hide where the problem lives.
2. Say points or per cent, and check the denominator.
3. Choose the chart for the comparison and label it honestly.
4. Round to what the data can bear, and name what is missing.

Exit task Write the most important number in your case, with its year, its source and one caveat.

Before the next session

Independent work

Finish your three charts. Ask your pod's evidence checker to review one chart for an honest scale and honest wording. About 50 minutes.

Read

Spiegelhalter, David. The Art of Statistics: Learning from Data. London: Pelican, 2019. Chapter 1.

Next

Day 13: Experience and comparison

Tomorrow: what people's experience, and other places' attempts, can tell us.